ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** break in the circuit	Loose contact stops current.
2	**(a)** heating	All use resistive heating.
3	**(d)** compass needle deflection	Oersted.
4	**(b)** nichrome	Higher resistance.
5	**(a)** the current is switched off	Electromagnet demagnetises.
6	**(d)** coin, key and iron nail	Metals.
7	**(d)** the switch is open	No current.
8	**(c)** glows brighter	Greater voltage.
9	**(b)** circles around the wire	Concentric circles.
10	**(d)** more clips	More turns, stronger.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fused filament or loose connection (1); Check connections and try another bulb (1).	1+1
12	More turns produce a stronger magnetic field (1); More force on iron pins (1).	1+1
13	Thin wire (1); It has higher resistance (1).	1+1
14	Copper has low resistance (1); Nichrome has high resistance, so it heats (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Broken connection or weak cell: replace it (1); No iron core: use an iron nail (1); Too few turns: add turns (1).	1+1+1
16	It switches off when current is too large (1); Using a magnet or heating strip (1); Can be reset (1).	1+1+1
17	Cord must carry current with little heat (1); Coil must produce heat (1); Nichrome has high resistance (1).	1+1+1
	OR: Thin wires have high resistance and overheat (1); Thick wires carry more current safely (1); Insulation prevents shocks (1).	1+1+1
18	Earth wire connects the metal body to the ground (1); Leaks current to earth (1); Prevents shocks (1).	1+1+1
	OR: Faulty or bare wires (1); Wet hands or floor (1); Overloading (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Set-up with wire, cell, switch, compass (1); Needle deflects when current flows (1); Reversing gives opposite deflection (1); Above and below the wire deflections are opposite (1); Conclusion: magnetic field around the wire (1).	1+1+1+1+1
	OR: Fuse and MCB: excess current (1); Earthing: leak current (1); Insulation: shocks (1); Switch on live wire (1); Proper plugs, no overloading (1).	1+1+1+1+1
20	Coil on soft iron core (1); Current magnetises the core (1); Armature attracted, hammer hits gong (1); Contact breaks, magnet loses power (1); Spring returns, cycle repeats (1).	1+1+1+1+1
	OR: Current in a large coil makes it a strong magnet (1); It lifts iron scrap (1); Switching off drops it (1); Strength can be changed (1); Permanent magnets can't be switched off (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** heats less due to low resistance	Low resistance.
2	**(a)** melt as 17 A exceeds 15 A	2+5+6+4 = 17 A.
3	**(a)** become very weak	No iron core to magnetise.
4	**(b)** north and south poles	Electromagnet has poles.
5	**(a)** 220 V	Household voltage.
6	**(d)** fuse wire	Low melting point.
7	**(a)** greater resistance	Thin means more resistance.
8	**(c)** parallel	Each gets same voltage.
9	**(a)** the fuse will melt	Current exceeds the rating.
10	**(a)** it may cause a shock	Safety.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Switching off cuts the live wire (1); Appliance becomes safe to touch (1).	1+1
12	A magnetic needle can turn freely (1); It shows the direction of the field (1).	1+1
13	All current flows through it (1); It can cut off the current (1).	1+1
14	They are in parallel (1); Each has an independent path (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Current passes through the tungsten filament (1); It heats to a high temperature (1); Glows white hot and gives light (1).	1+1+1
16	Thin wire has more resistance (1); Less current flows (1); Hence the bulb glows dim (1).	1+1+1
17	Leaving lights and fans on (1); Use energy-saving LED bulbs (1); Switch off when not required (1).	1+1+1
	OR: Bulb still glows (1); Current direction reverses (1); Compass needle deflects in the opposite direction (1).	1+1+1
18	Series: single path; simple (1); Parallel: many paths; each bulb works independently (1); Homes use parallel (1).	1+1+1
	OR: (a) Electrical to heat (1); (b) Electrical to sound (1); (c) Chemical to electrical (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total = 3+2+4+3 = 12 A (1); Exceeds 10 A (1); Fuse melts and breaks the circuit (1); Avoid overloading (1); Use a higher-rated circuit only with proper wires (1).	1+1+1+1+1
	OR: Coin: bulb glows, metal conductor (1); Eraser: does not glow (1); Salt water: glows (dim) as ions conduct (1); Pencil lead (graphite): glows, conductor (1); Material decides conduction (1).	1+1+1+1+1
20	Heating: heater, iron (1); Magnetic: bell, crane (1); Heating precaution: keep away from flammable items (1); Magnetic precaution: use insulated wires (1); Both depend on current flowing (1).	1+1+1+1+1
	OR: More current, more heat (1); Thin wire has more resistance (1); More heat in thin wire (1); Use thick wires for supply, thin for heating (1); Overheating can cause fire (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** electric bell	Bell uses magnetism.
2	**(b)** reset after tripping	Reusable.
3	**(d)** a shock	Current passes through the body.
4	**(c)** gets magnetised	Electromagnet.
5	**(d)** deflected	Magnetic effect.
6	**(a)** tin-lead alloy	Low melting point.
7	**(c)** get interchanged	Direction of current reversed.
8	**(c)** not work well because the current short-circuits	Current goes through turns directly.
9	**(d)** also go off	The circuit breaks.
10	**(d)** loses magnetism quickly when current stops	Quick demagnetisation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To add their voltages (1); Gives larger voltage to the circuit (1).	1+1
12	Reversing current reverses deflection (1); Shows field direction changes (1).	1+1
13	It will not melt at the right time (1); The appliance and wires may overheat (1).	1+1
14	Load can be dropped by switching off the current (1); Permanent magnets would hold on (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iron becomes magnetised and increases the field (1); Makes the magnet strong (1); Without it, the magnet would be weak (1).	1+1+1
16	Keep the compass near the wire (1); Keep away from other magnets and iron (1); Do not leave the current on for long (1).	1+1+1
17	Thick wire lets more current pass (1); More current means stronger magnet (1); Too thick wire gives fewer turns (1).	1+1+1
	OR: Fuse wire melts at a low temperature (1); Copper has low resistance, so less heat (1); Hence the fuse breaks first (1).	1+1+1
18	An electromagnet (1); The current magnetises the nail (1); Switching off removes magnetism (1).	1+1+1
	OR: Rubber is an insulator (1); It blocks current (1); Protects from shock (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Nichrome: high resistance, high melting point (1); Tungsten: very high melting point (1); Tin-lead alloy: low melting point (1); Chosen based on use (1); Each property is vital for safe function (1).	1+1+1+1+1
	OR: Large total current (1); Wire gets hot (1); Fire hazard (1); Spread loads over several circuits (1); Use proper fuse or MCB (1).	1+1+1+1+1
20	Core: soft iron boosts field (1); Turns: more turns, stronger field (1); Current: more cells increases current (1); Compare by number of pins lifted (1); Keep other things constant for a fair test (1).	1+1+1+1+1
	OR: Nichrome coil inside (1); Current heats the coil (1); Heat passes to the metal plate (1); Thermostat controls temperature (1); Heating effect (1).	1+1+1+1+1

Mistakes to avoid:

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- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.

- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
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